

DATA 298A — Frequently Asked Questions

Fall 2026. Read alongside the DATA 298A introduction deck, the demo schedule document, and the project list for your section (11, 21 or 31). Where those documents and this one appear to disagree, the introduction deck governs — tell me and I will correct this file.

1. Choosing and changing your topic

Q1. Where do project topics come from?

From the project list for your section. Sections 11, 21 and 31 each have their own deck of 20 topics. Every topic is scoped for four graduate students across two semesters and names the models, datasets and evaluation plan you are expected to work from.

Q2. Can two teams take the same topic?

No. One team, one topic, first come first served within your section. Email your choice; the topic is not yours until you receive written confirmation.

Q3. Can we take a topic from another section's list?

Ask first. The lists are balanced so each section covers the same range of areas, and moving a topic between sections can create a duplicate. If the topic is unclaimed in both sections it will usually be approved.

Q4. Can we propose our own topic?

Yes, with written approval before you start. A proposal must clear the same bar as the topics on the list:

- A problem that is genuinely open, not a tutorial reimplementations.
- An end-to-end system, with the architecture sketched.
- Four or more distinct models you will build and compare.
- A named benchmark or dataset, and the metrics you will report.
- A realistic compute budget and a data-access plan.
- Enough work to occupy four people for two semesters.

A proposal that is a wrapper around one API call will not be approved.

Q5. Can we change topics after we have started?

Only before the abstract is due on 9/9, and only with approval. After the Abstract Presentation you are committed. If your topic turns out to be infeasible for a reason you could not have foreseen — a dataset withdrawn, hardware you cannot get — raise it immediately rather than at the midterm.

Q6. Some topics are marked [YC]. What does that mean?

Those topics are drawn from Y Combinator's Fall 2026 Requests for Startups, which describes problems investors currently consider unsolved. The tag carries no grading advantage. It tells you the problem is live.

Q7. Our topic needs GPUs we do not have. What do we do?

Raise it at Checkpoint 1 on 9/10, under risks. Options include SJSU cluster access, cloud credits, using a smaller model class for development and scaling once, or substituting a model from your topic slide for a cheaper one with approval. What is not acceptable is discovering the problem in November.

2. Teams, Linear, and individual contribution

Q8. Do we need to give the instructor and the ISA access to our project tracking tool?

Yes. This replaces the earlier answer in this FAQ, which said screenshots were sufficient. They are no longer sufficient.

- Create one Linear team for your project, named after your topic.
- Invite the instructor and the ISA (shristi.kumar@sjsu.edu) with admin access.
- Create a Linear Project for the semester and enter the course milestones.
- Turn on two-week Cycles, aligned to the milestone schedule in the introduction deck.

This must be done by Thursday 9/3 and is checked at the Abstract Presentation on 9/10.

Q9. What has to be in Linear?

Every task, as an issue with:

- One assignee — never a group, never unassigned.
- An estimate, a cycle, and a label: research / data / model / infra / eval / writing.
- A link to the artifact it produced: a pull request, commit, dataset version or experiment run.

Workflow is Backlog → Todo → In Progress → In Review → Done. Nothing moves to Done without a linked artifact.

Q10. How are individual contributions graded?

From the Linear record, read against the Git history and the demo. Those three should agree. At each checkpoint we look at issues completed per member per cycle, whether they carry linked artifacts, balance across labels, and whether the author of the linked pull request matches the assignee.

Individual grades can differ inside one team. A strong project does not earn everyone the same mark.

Q11. What if work is not tracked in Linear?

For grading purposes it did not happen. This is not a technicality — it is the only evidence of who did what that survives the semester.

Q12. One of our teammates is not contributing. What do we do?

Raise it at the next checkpoint, in the meeting, with the Linear board open. It can be fixed on 10/8 or 11/12. It cannot be fixed on 12/8, and a complaint that arrives in the final week with no Linear record behind it is very difficult to act on.

Q13. Can we choose our own team members?

Generally no. In the real world you rarely choose your team; you are assigned one. Be prepared to work with whoever you are teamed with.

Q14. How do we email a question?

Put the course number, section number and project topic ID in the subject line, and copy every member of your team. Emails without this are answered last.

3. Demos and presentations

Q15. What are the graded sessions this semester, and when?

Four, on the official course dates below. The shaded rows are the graded presentations; everything else is a submission deadline.

Date	Day	Activity or deliverable
8/27	Thu	First online meeting — course overview, teams, topic lists released
9/3	Thu	Topic confirmed by email; Linear board created and ISA invited
9/9	Wed	Project Abstract due
9/10	Thu	Candidacy Form due
9/10	Thu	CHECKPOINT 1 — Abstract Presentation, 9:00–11:45 am
9/24	Thu	Progress Report 1 due
10/7	Wed	Workbook 1 due
10/8	Thu	CHECKPOINT 2 — Team Meeting 1, 9:00–11:45 am
10/22	Thu	Progress Report 2 due
11/11	Wed	Workbook 2 due
11/12	Thu	CHECKPOINT 3 — Team Meeting 2, 9:00–11:45 am
12/7	Mon	Final Report due
12/7	Mon	Project Presentation slides due
12/8	Tue	CHECKPOINT 4 — Project Presentation (Conference Day), 9:00 am–12:00 pm

Q16. What must be true by each checkpoint?

- **Thu 9/10 — Abstract Presentation (Professor):** Topic confirmed, abstract submitted 9/9, GitHub repo live, Linear board created with the ISA invited, baseline paper identified
- **Thu 10/8 — Team Meeting 1 (Professor and ISA (two presentations)):** Clean dataset, fixed splits, metrics chosen and justified, end-to-end data extraction and cleaning demonstrated
- **Thu 11/12 — Team Meeting 2 (Professor and ISA (two presentations)):** Automated cloud pipeline, working prototype demonstrated live, published baseline reproduced, first model trained
- **Tue 12/8 — Project Presentation (Conference Day) (Professor and ISA):** Benchmark table with error bars, one ablation, 50+ categorized failures, reproducibility check, 298B plan

Q17. Earlier drafts of this FAQ referred to six demos in weeks 3, 5, 8, 9, 12 and 15. Which is right?

The four dates above. An earlier revision used a generic 15-week outline before the official schedule was published; that outline is superseded. The content did not disappear — it was folded into the four sessions:

- Topic, abstract, initial state of the art and project setup are Checkpoint 1 (9/10).

- Requirements, literature survey and the data pipeline are Checkpoint 2 (10/8), split across a Professor presentation and an ISA demonstration.
- Model proposals, the cloud pipeline, the working prototype and the baseline reproduction are Checkpoint 3 (11/12), again split across two presentations.
- The final prototype, benchmark, ablation, failure study, reproducibility check and 298B plan are Checkpoint 4 (12/8).

Q18. What is the presentation format?

Eight slides or fewer, 17 minutes of presentation and 3 minutes of questions, per audience. The final Project Presentation is the exception: up to 11 slides, because it has to cover the benchmark, the ablation, the failure study and the live prototype. A slide deck is expected at every presentation.

Q19. Who presents?

All four members attend and each member presents the part they did. A member who does not present does not receive credit for that demo.

Q20. What do we submit before a demo?

The GitHub repository link at the demo commit, the Linear board link, and a one-to-two page demo memo.

Q21. Can we postpone a demo?

No. There is no postponement, and a missed demo scores zero.

Q22. Can a demo be a recording or screenshots instead of running live?

Demos 1, 2 and 3 may use static artifacts where a service is genuinely unavailable, provided you say so. Demos 4 and 6 must run live — those are the prototype checkpoints, and the audience chooses the input. If any part of a demo is mocked or hardcoded, say so out loud before you run it.

4. Models, benchmarking and evaluation

Q23. What counts as one of our four models?

A model your team trained, fine-tuned or adapted. Four genuinely different modelling approaches — not four hyperparameter settings of the same model. Your topic slide names four or more candidates; treat them as the floor, not the ceiling.

Q24. Does the baseline we reproduce count as one of the four models?

No. It is separate, and it is doing a different job.

The four models are the approaches your team builds and compares against each other. The reproduced baseline is a published number you regenerate on your own benchmark — the reference point all four are measured against, and the evidence that your evaluation setup is sound. So the arithmetic is:

- 298A: one model you built, plus the reproduced baseline.
- 298B: the three remaining models.
- Four models in total, all built by you, all compared against that one baseline.

If your topic slide lists something described as a baseline or a bound among its named models, you still build it — it is one of your four. The reproduced published result is in addition to those.

Q25. Can we use OpenAI or another hosted API as one of our models?

It can be one of the models for comparison, but it cannot be your main model, because calling a hosted endpoint is not building a model. You still need four distinct models you built, adapted or reproduced.

Q26. What is an evaluation metric, exactly?

A number that measures how well the system does the task, computed on data the model has not been trained on, using a procedure you fixed before you saw any results. The right metrics depend on your topic:

Topic type	Report at minimum
Classification / detection	Accuracy, precision, recall, F1, confusion matrix, PR curve; subgroup breakdown where people are involved
Ranking / retrieval / search	Recall@k, NDCG, MRR; latency at a realistic serving configuration
Generation / LLM systems	Task-specific accuracy, plus a human or LLM judge with reported inter-rater agreement; cost per query
Robotics / control / RL	Success rate, SPL or equivalent, sample efficiency, robustness under perturbation
Systems / infrastructure	Throughput, latency percentiles, cost per unit of work, quality delta versus the unoptimized baseline
Security / safety	Detection rate and false-positive rate at a stated operating point, and generalization to unseen attacks

Your topic slide is authoritative. Choose your metrics for Team Meeting 1 on 10/8 and justify them, before you have results.

Q27. What is a baseline, and why must we reproduce one?

A published result from the literature, reproduced by you, on the same benchmark. Without it, a comparison means nothing: beating a baseline you implemented badly is not evidence. Identify the baseline at Checkpoint 1 on 9/10 with the exact table or figure its number comes from, work on the reproduction through October, and present it at Team Meeting 2 on 11/12.

Q28. What if we cannot reproduce the published number?

Present that as a finding, with evidence of what you tried. A documented failure to reproduce is a legitimate and often valuable result. An undocumented one is a missing deliverable.

Q29. Is one training run enough to report a result?

No. Report multiple seeds with error bars and state the number of runs. A single lucky run is not a result.

Q30. What is an ablation?

You remove one component of your system and report the change in your headline metric. If removing a component changes nothing, report that too — it is a finding, and often a more interesting one than a small win.

Q31. What is the failure study, and how big does it need to be?

At least 50 failure cases sampled from your evaluation runs, sorted into a taxonomy you define, with the distribution across categories reported and the top three causes named. You also state the conditions under which your system should not be used. It is due at the final presentation on 12/8 and is graded as heavily as accuracy.

Q32. What are the data requirements for LLM projects?

You will not train an LLM from scratch. You fine-tune or adapt a pre-trained model, so the data requirement you document is the fine-tuning or evaluation data, not a pre-training corpus.

5. Prototype, pipeline and reproducibility

Q33. Does 298A require implementation, or is it a planning semester?

It requires implementation. 298A is not a planning semester — you finish it with something that runs.

The six deliverables are: a deployed working prototype; at least one model your team built; a running evaluation harness; a reproduced published baseline; a first failure study; and the 298B plan.

Q34. What counts as a working prototype?

It counts if:

- It runs end to end and produces a real, scored output.
- It is deployed in the cloud, with code in GitHub and a pinned environment.
- Another team could clone it and run it from your README without asking you a question.
- It contains at least one model your team trained, fine-tuned or adapted.

It does not count if:

- It is a notebook that only runs on one laptop.
- It is a set of screenshots of something that used to work.
- The interface is real but the outputs are hardcoded.
- It is an API call with a user interface on top and no model of your own.
- It only works on the inputs you rehearsed.

A thin slice that genuinely works beats a broad system that only works in the demo. Build the thin slice well before 11/12.

Q35. What does the data pipeline have to include?

Ingestion from source, transformation and cleaning, storage in a warehouse or lakehouse (BigQuery, Redshift, Snowflake, S3, GCS, Delta Lake), and monitoring and management (Airflow or Cloud Composer — logs, status, retries). It must be automated. A pipeline someone runs by hand is not automated, and you should be able to show a real log from a real run.

Q36. What is the reproducibility check?

At the final presentation on 12/8, another team clones your repository and runs it without your help, then reports to the class what happened. Pinned environment, seeds, logged experiments and

documented hardware are what make this survivable. Any result you cannot regenerate on request is treated as not obtained.

6. Literature survey, abstract and references

Q37. How much literature survey is expected?

Enough that your team can say what the important papers in your area are and what the state of the art is. There is no fixed number, but roughly 30 to 50 papers is the usual range across the project.

Checkpoint 1 asks for 8 to 12 reviewed in depth, including the one baseline you will reproduce.

Q38. What should the abstract contain?

The four headings from the department template, in order: Purposes, Tasks, Outcomes, Applications. Ideally 200 to 250 words, and no more than 400. Use APA format. List project members in alphabetical order, not in order of contribution. When you present it, put each component — introduction, importance, goal, issues, methodology, expected outcome — on its own slide.

Q39. Anything the department abstract template does not say to include?

Two things the course adds. Under Tasks, name the four models your team will build and the published baseline you will reproduce — the baseline is not one of the four. Under Outcomes, state the technical merit as a number on a named benchmark or split, not as an adjective. Both go in the declarations table that is submitted with the abstract and does not count toward the word limit.

Q40. Is there a sample abstract?

No.

Q41. Can we cite Medium, Forbes, or similar sources?

Journal and conference papers are strongly preferred. Other articles may be cited, but they should not exceed 20% of your references.

Q42. APA or IEEE?

APA is recommended. IEEE is also acceptable. Pick one and be consistent.

7. Reports, workbooks and progress reports

Q43. What are the deliverables for this semester?

- A complete report in the project template.
- A full data pipeline through to the data warehouse.
- A deployed working prototype demonstrating feasibility.
- A benchmark table with your first model against a reproduced baseline.
- A failure study of at least 50 categorized cases.
- The 298B plan: which three remaining models you will build, on which benchmark, against the baseline you reproduced.

Q44. What is expected in Workbook 1?

Report sections 1 and 2, written as report text. Workbook 1 uses the report headings — 1.1 Project Background and Executive Summary through 1.5 Literature Survey, then 2.1 Data Management Plan through 2.4 Project Schedule, including all eight parts of 2.2, then References. It is not a separate exercise and it is not a set of notes: it is your part of the final project report, submitted early so it can be corrected while there is still time.

Q45. What is expected in Workbook 2?

Report sections 3 and 4, on the same rule. Section 3 is 3.1 Data Process through 3.6 Data Statistics; section 4 is 4.1 Model Proposals through 4.4 Model Evaluation Methods. Leave 4.5 and 4.6 as headings — the first model and the reproduced baseline are written up in the final report. Do not rewrite sections 1 and 2 in Workbook 2; record any revision in the carry-forward table instead.

Q46. If the workbook is already report text, what is left for the final report?

Sections 5 through 8 and the appendices, plus the revisions to sections 1–4 that come out of the two Team Meetings. Teams that treat the workbooks as notes end up writing the whole report in the last two weeks; teams that treat them as report text arrive at 12/7 with roughly half of it already graded and corrected.

Q47. What do the four columns in Part A of the progress report mean?

"Planning" is how much of the thinking and scoping for that section is done. "Investigation" is how much of the research, reading and experimentation is done. "Complete" is how much of the actual work is finished. "Document" is how much of the written report for that section is finished. They are usually four different numbers, and a section that is 90% complete and 0% documented is a warning sign. Sections you have not started are still listed, at 0%.

Q48. The Part A row numbering does not match my old progress report handout.

Correct — the handout predates the current project report template. Section 2 is now 2.1, 2.2 with its eight parts, 2.3 and 2.4. The old handout also had 3.4 and 3.5 swapped: it is 3.4 Data Transformation, then 3.5 Data Preparation. Rows for sections 4 through 8 have been added so the same table works for both progress reports. Use the numbering in the template.

Q49. What goes in Part B, Minutes of Meetings?

One block per team meeting held during the reporting period: time and date, location or media, participants, then a row per report section with achievements, issues and recommended solutions. Copy the block as many times as you need. These are your own team meetings, not the graded Team Meetings with the instructor and ISA.

Q50. Do we have to keep the progress report format?

Yes. Parts A and B are the department format and must be retained exactly. Parts C and D — the Linear work summary and the blocker register — are course additions and follow after them; do not fold them into Parts A or B.

Q51. Is there a presentation for the progress report?

No.

8. Generative AI policy

Q52. Can I generate my project report using GenAI (ChatGPT, Copilot, Gemini or similar)?

No. You may not use GenAI tools to generate any part of your project report. The report must be written by you and your team members.

If there is sufficient evidence that a report, or a substantial portion of it (more than 10%), was generated by GenAI, the team receives an automatic F, and it may be treated as an academic integrity violation under university policy.

Q53. How do you check for GenAI-generated reports?

All reports are run through:

- Turnitin plagiarism review and AI writing detection (Canvas).
- GenAI detection tools including GPTZero, Copyleaks AI Content Detector and Originality.ai.

These tools are not used in isolation. They are one part of the review. We also look for the usual signs of AI-generated writing: a mismatch with your previous writing, text that is fluent but generic, claims with no specific evidence behind them, and an inability to answer questions about your own report.

Q54. Can we use GenAI to help write code?

Yes, for code, subject to the same rule that applies to every other tool: you must understand what you submitted and be able to explain it when asked at a demo. The prohibition is on generating the written report.

9. Grading, deadlines and 298B

Q55. What do "Poor", "Good" and "Excellent" mean in the rubrics?

Every rubric section is rated on three levels. The rating is per section, not per report — the same report can be excellent in one section and poor in another.

- **"Poor"** — The section is missing, incomplete, or does not meet the stated expectation. Something required is absent, or present but wrong.
- **"Good"** — The section meets the expectation with no major flaws. This is the target, and a report rated good throughout is a successful report.
- **"Excellent"** — The section goes beyond what was asked, in a way a reader can point to. Rare by design.

Most sections should be good. Excellent is what a small number of sections earn, and it is not required to succeed in this course.

Q56. What actually moves a section from "Good" to "Excellent"?

Concretely, in this course:

- You reproduced the published baseline, and explained the gap instead of glossing over it.
- Results carry error bars over multiple seeds, and the number of runs is stated.
- An ablation changed your conclusion — and you said so.
- The failure study found something you did not expect, and you acted on it.

- Another team ran your repository without asking you a question.
- A reader outside your team understands the contribution in one pass.

None of these are about volume. A longer report is not a better one.

Q57. Can we have a deadline extension?

No.

Q58. What are the most common reasons teams lose marks?

- The prototype only works on the rehearsed inputs.
- Metrics were chosen after the results were seen.
- The baseline was implemented by the team, badly, and then beaten.
- One seed, no error bars, and a small improvement claimed as a result.
- The failure study is a paragraph rather than 50 categorized cases.
- Linear was filled in the night before the presentation.
- One member did most of the work and the team hoped nobody would ask.
- The report is fluent, general, and clearly not written by the team.

None of these are difficult to avoid. All of them are easy to detect.

Q59. What are we doing in 298B?

The same project, the same repository and the same Linear board. Nothing restarts.

- Complete the four models. The three remaining models from your topic slide, built and compared against the baseline you reproduced in 298A — same benchmark, same fixed splits, same metrics.
- Retrain, do not just train. The pipeline must be able to retrain a model on new data without hand-holding — on a schedule or on drift — with versioned datasets and model artifacts, an evaluation gate before promotion, and rollback when the new model is worse.
- The 298A prototype becomes a complete application with a real user interface on top of the pipeline, deployed and usable by someone outside the team.
- The final benchmark: all four models measured and compared, with error bars over multiple seeds, ablations, and an expanded failure study.
- Analysis and interpretation — why the results came out as they did. A table with no interpretation is not a finding.
- A paper-quality write-up, and publication where the work supports it.

Q60. Do we keep using Linear in 298B?

Yes, and the requirement tightens rather than relaxes.

- Use the same board. Do not create a new one. It stays open across the winter break and the instructor and ISA remain admins.
- Cycles resume in the first week of 298B.
- The 298B plan you presented on 12/8 — the three remaining models, the benchmark, the compute budget, the schedule — is entered as Linear milestones with issues assigned to named people before week 2.

- Individual grading works the same way: issues completed per member per cycle, each with a linked artifact, read against the Git history and the demo.
- Anything left undone in 298A stays in the backlog with its original creation date. It does not quietly disappear over the break.

A clean 298A board is the cheapest thing you can hand your February self.

Q61. What does "retrain" mean in 298B, and how is it different from training?

Training happens once, by hand, on the data you had. Retraining is the system doing it again, by itself, when the world changes. In 298B the pipeline must support:

- A trigger — a schedule, a data-volume threshold, or a drift detector that fires when incoming data no longer resembles your training distribution.
- Versioned inputs and outputs — the exact dataset version and the exact model artifact, so any deployed prediction can be traced back to what produced it.
- An evaluation gate — the retrained model is scored on your fixed test split and is only promoted if it beats the model in production.
- Rollback — if the new model is worse, or worse for one subgroup, you can return to the previous one quickly.

Demonstrate it: change the data, let the pipeline retrain, and show the gate either promoting or rejecting the result.

Q62. How much of 298A can we defer to 298B?

Very little, and it is a bad trade. The 298A deliverables — a deployed prototype, one model you built, a running evaluation harness, a reproduced baseline, a failure study and the 298B plan — are the foundation 298B is built on. The teams that struggle in 298B are the ones that arrive without a working prototype.